

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular & Supplementary Winter Examination-2023****Course: B. Tech.****Semester : III****Branch :** Electronics and Computer Engineering/ Electronics and Computer Science Engineering**Subject Code & Name:** Programming, Data Structure and Algorithm using C (BTECPC303)**Max Marks: 60****Date:06-01-24****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is Linked List? Explain various operations performed on Linked List.	L2/CO1	6
B) Draw a labeled diagram of linked list having 5 nodes; a)Singly linked list b)Singly Circular linked list c)Doubly Circular linked list	L2/CO1	6
C) Write a C program to delete first node of singly linked list.	L1/CO1	6
Q.2 Solve Any Two of the following.		12
A) Define Stack. Explain working of stack with diagram.	L3/CO2	6
B) Explain Infix, Prefix and Postfix expression.	L2/CO2	6
C) Write a C program to implementation of stack using linked list	L3/CO2	6
Q. 3 Solve Any Two of the following.		12
A) Define Queue. Explain working of queue with diagram.	L2/CO3	6
B) What is the limitation of simple queue? Explain circular queue.	L3/CO3	6
C) Write a note on i) PRIORITY QUEUE ii)DEQUEUE	L2/CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain the linked list representation of Binary Tree	L2/CO4	6
B) Define i) Tree ii) Degree of Node iii) Degree of Tree iv) Graph v) Adjacent Node vi) Weighted Graph	L2/CO4	6
C) State various ways of graph representation. Explain anyone.	L2/CO4	6
Q. 5 Solve Any Two of the following.		12
A) What is Algorithm? Define Asymptotic notation with example.	L2/CO5	6
B) Explain Binary Search with help of following array. int array[8] = {8,1,3,7,39,27,16,45}	L3/CO5	6
C) Write an algorithm for selection sort.	L2/CO5	6

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